

广义积分

证: $F'(x) = f(x)$, $\forall b > a$ $\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$

$$b \rightarrow +\infty \quad \int_a^b f(x) dx = \lim_{b \rightarrow +\infty} F(b) - F(a) = F(+\infty) - F(a)$$

例: 讨论广义积分 $\int_1^{+\infty} \frac{1}{x^p} dx$ 的收敛性

$$\int \frac{1}{x^p} dx = \begin{cases} -\frac{1}{p-1} \cdot x^{-p+1} & p \neq 1 \\ \ln |x| & p = 1 \end{cases}$$

$p > 1$ 时 $\int_1^{+\infty} \frac{dx}{x^p} = \left. \frac{-1}{(p-1)x^{p-1}} \right|_1^{+\infty} = \frac{1}{p-1}$

$p = 1$ 时 $\int_1^{+\infty} \frac{dx}{x^p} = \int_1^{+\infty} \frac{dx}{x} = \lim_{x \rightarrow +\infty} \ln x \Big|_1^{+\infty} = +\infty$

$p < 1$ 时 $\int_1^{+\infty} \frac{dx}{x^p} = \left. \frac{1}{1-p} x^{1-p} \right|_1^{+\infty} = \frac{1}{1-p} [\lim_{x \rightarrow +\infty} x^{1-p} - 1] = +\infty$

例: $\int_{-\infty}^{+\infty} \frac{1}{x^2+2} dx$

$$= \int_{-\infty}^{+\infty} \frac{\frac{\sqrt{2}}{2} d\frac{x}{\sqrt{2}}}{2(1+(\frac{x}{\sqrt{2}})^2)} = \frac{1}{\sqrt{2}} \arctan \frac{x}{\sqrt{2}} \Big|_{-\infty}^{+\infty}$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\pi}{2} \cdot 2 = \frac{\pi}{\sqrt{2}}$$

广义分部积分公式 $u(x), v(x)$ 在 $[a, +\infty)$ 上连续可导. $+\infty$ 是其唯一奇点

$$\int_a^{+\infty} u(x) dv(x) = u(x) v(x) \Big|_a^{+\infty} - \int_a^{+\infty} v(x) u'(x) dx$$

例: $\int_0^{+\infty} e^{-x} \cos x dx = \int_0^{+\infty} \cos x d(-e^{-x})$

$$= -e^{-x} \cos x \Big|_0^{+\infty} - \int_0^{+\infty} e^{-x} \sin x dx$$

$$= 1 + e^{-x} \sin x \Big|_0^{+\infty} - \int_0^{+\infty} e^{-x} \cos x dx$$

$$I = \frac{1}{2}$$

例: 求 $\int_0^{+\infty} (\frac{1}{1+x} - \frac{x^2}{1+x^3}) dx$

$$= \lim_{b \rightarrow +\infty} \int_0^b (\frac{1}{1+x} - \frac{x^2}{1+x^3}) dx = \lim_{b \rightarrow +\infty} (\ln(1+x) \Big|_0^b - \frac{1}{3} \ln(1+x^3) \Big|_0^b) = \lim_{b \rightarrow \infty} \ln \frac{1+b}{\sqrt[3]{1+b^3}} = 0$$

$$\int_1^{+\infty} \frac{1}{x\sqrt{x}} dx$$

收敛

$$\int_0^1 \frac{1}{x\sqrt{x}} dx$$

发散

$$\int_0^1 \frac{1}{\sqrt{1-x}} dx$$

收敛

$$\int_0^{+\infty} \frac{1}{x\sqrt{x}} dx$$

发散

$$\int_1^2 \frac{1}{(2-x)^3} dx$$

发散

例: $\int_1^2 \frac{1}{x^3 \sqrt{4-x^2}} dx$

例变量代换 令 $t = \frac{1}{x}$

$$\begin{aligned} \text{原式} &= \int_1^{\frac{1}{2}} \frac{1}{\frac{1}{t^3} \sqrt{4-\frac{1}{t^2}}} \cdot (-\frac{1}{t^2}) dt = \int_{\frac{1}{2}}^1 \frac{t}{\sqrt{4t^2-1}} dt = \frac{1}{8} \int_{\frac{1}{2}}^1 \frac{8t}{\sqrt{4t^2-1}} dt = \frac{1}{8} \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{4t^2-1}} d(4t^2-1) \\ &= \frac{1}{8} \cdot 2 \sqrt{4t^2-1} \Big|_{\frac{1}{2}}^1 = \frac{\sqrt{3}}{4} \end{aligned}$$

例: 计算广义积分 $\int_0^{+\infty} \frac{1}{1+x^4} dx$

令 $x = \frac{1}{t}$

$$\Rightarrow I = \int_{+\infty}^0 \frac{1}{1+\frac{1}{t^4}} (-\frac{1}{t^2}) dt = \int_0^{+\infty} \frac{t^2}{1+t^4} dt = \int_0^{+\infty} \frac{x^2}{1+x^4} dx$$

$$I = \frac{1}{2} \int_0^{+\infty} \frac{1+x^2}{1+x^4} dx = \frac{1}{2} \int_0^{+\infty} \frac{1+\frac{1}{x^2}}{\frac{1}{x^2}+x^2} dx = \frac{1}{2} \int_0^{+\infty} \frac{1}{2+(x-\frac{1}{x})^2} d(x-\frac{1}{x}) = \frac{1}{2\sqrt{2}} \arctan \frac{x-\frac{1}{x}}{\sqrt{2}} \Big|_0^{+\infty} = \frac{\pi}{2\sqrt{2}}$$